DOCUMENT 1 OF 3 · PRODUCT REQUIREMENTS

Post-Operative Remote Patient Monitoring

Product Requirements Document. MVP scoped to total knee replacement, single clinic, India.

Version 0.1 · Draft for clinical and engineering review

Companion to the High Level and Low Level Design documents

14 September 2026

SECTION 1

Overview and problem

Between the day a knee replacement patient goes home and their first follow-up appointment, nobody is watching. A meaningful share of readmissions in that window trace back to complications that would have been an easy catch if someone had been.

This product closes that gap. After discharge, the patient completes a short daily check-in on their phone. A rules engine evaluates each submission against thresholds tuned to their own discharge baseline and their post-operative day. Anything concerning surfaces on a clinic dashboard sorted by urgency, and genuinely critical findings interrupt an on-call nurse immediately rather than waiting to be noticed.

Why knee replacement first

Recovery is predictable, milestones are well defined and easy for a patient to self-report, and the complication profile is well understood and arrives in a known order. The clinical logic is therefore tractable, and a surgeon can meaningfully approve threshold numbers. The architecture is procedure-agnostic; the procedure lives in configuration, not code.

Positioning

The Indian market is served by hardware-first vendors (contactless and connected vitals devices) and service-first providers (nurse home visits). No clear domestic equivalent exists to the international digital surgical-episode platforms. The opportunity is a lightweight patient-facing application layer that a hospital can adopt on top of whatever vitals hardware it already owns.

> **CORE PRODUCT BET**
>
> The highest-value feature is not data collection. It is a nurse queue that sorts itself by risk, so that attention lands on the right patient without anyone having to decide where to look.

SECTION 2

Goals and non-goals

Goals

ID	Goal
G-1	Detect post-discharge complications earlier than the scheduled follow-up appointment would.
G-2	Present the care team a single queue ordered by clinical urgency, requiring no triage decision from the user.
G-3	Guarantee that no critical finding goes unacknowledged, through automatic escalation.
G-4	Keep daily patient burden low enough that a 70-year-old completes the check-in unassisted on most days.
G-5	Make clinical thresholds legible and adjustable by a surgeon without a code change.

Non-goals for the MVP

ID	Non-goal	Rationale
NG-1	EHR or hospital information system integration	Procurement-bound. Replaced by manual enrolment.
NG-2	Automated analysis of wound photographs	Research problem. Photos are stored and reviewed by a human.
NG-3	Multi-tenancy and multi-procedure support	One clinic, one procedure, one active ruleset.
NG-4	In-product video consultation	Same-day review routes to the clinic's existing booking process.
NG-5	Formal patient-reported outcome instruments	Belong at six weeks and three months as outcome measures, not in a daily safety loop.
NG-6	Patient-visible risk scores	Generates anxiety and calls, not better outcomes.

SECTION 3

Users

Persona	Context and needs
Patient	Typically 55 to 75, on a low-to-mid-range Android handset, recovering at home and often in discomfort. May be assisted by a family member, who may complete the check-in on their behalf. Needs large touch targets, unambiguous questions in their own language, and no jargon. Must never be shown a risk score.
Nurse	Primary daily user. Manages a cohort of discharged patients alongside other duties, usually at a desktop. Needs to know who to attend to first, what specifically is wrong, and to close each item with a recorded outcome.
Surgeon	Approves the threshold configuration at programme level and adjusts it per patient at discharge. Receives escalations when a critical alert goes unacknowledged. Decides on same-day review.
Family member	Secondary, unregistered. May operate the app on the patient's behalf. Question wording must work in both the first and third person.

Operating assumption

The system is only as good as the person rostered to answer its alerts. A named on-call nurse with a defined coverage window is a deployment prerequisite, not a feature.

SECTION 4

Clinical basis

The risks after knee replacement arrive in a predictable order. The monitoring content and its cadence follow that order directly.

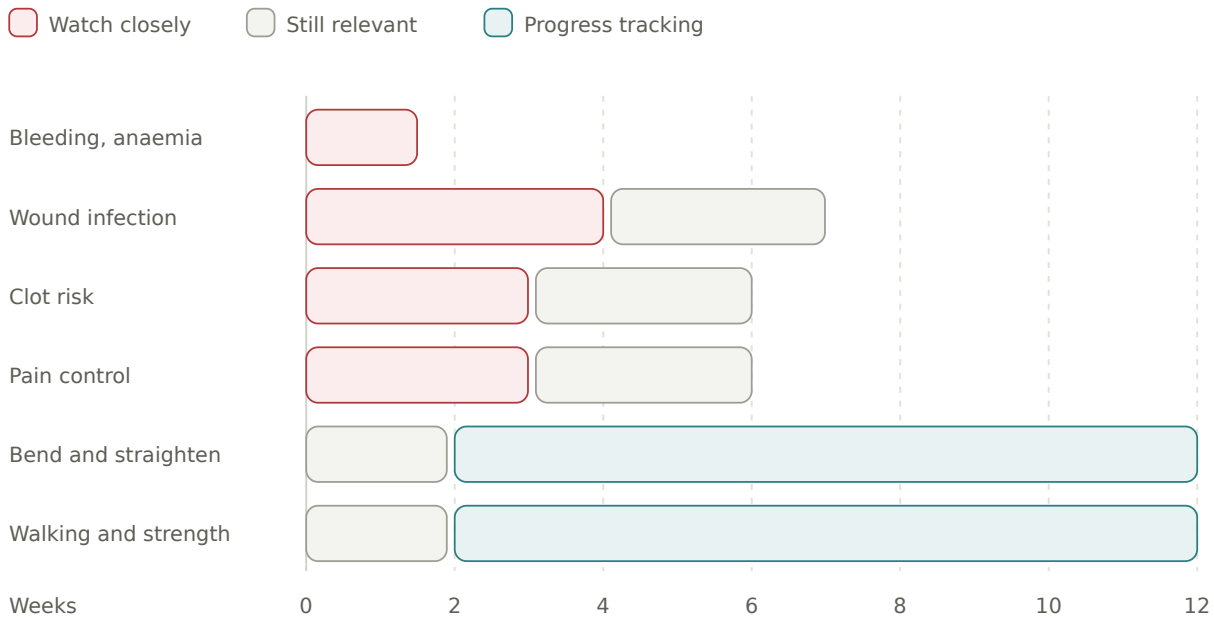


Figure 1. Risk windows after total knee replacement, driving both what is collected and when.

Two layers, one form

The product is two things that share a single check-in screen. A **safety layer** watching for complications, running daily, alert-driven, concentrated in the first three weeks. A **recovery layer** tracking progress, running weekly, milestone-driven, spanning the full twelve weeks. They have different cadences, thresholds and readers.

Design principle: trajectory over absolutes

Almost every complication announces itself as a break in an improving trajectory rather than as a single bad number. Pain of 7 on day two is expected; pain of 7 on day nine in a patient who was at 3 on day six is the signal. All thresholds are evaluated against the patient's own discharge baseline and recent history, and bands are indexed by post-operative day.

> **CLINICAL GOVERNANCE**
>
> Every threshold value, band boundary and milestone in this document is a starting point requiring surgeon approval before any patient use. Values vary by surgeon, implant and institution. The system must record which approved ruleset version produced each decision.

SECTION 5

Functional requirements

5.1 Enrolment

ID	Requirement
FR-1	A clinician can enrol a patient by entering name, phone number, preferred language, surgery date and discharge date.
FR-2	Enrolment captures the discharge baseline: temperature, pulse, oxygen saturation, pain score, knee bend, walking distance and walking aid. These become the comparison anchor for all later readings.
FR-3	Enrolment records explicit, versioned patient consent for health data processing, with timestamp and consent text version.
FR-4	Enrolment assigns an approved ruleset version to the patient. A surgeon may override individual thresholds for that patient.
FR-5	The patient receives an enrolment link by SMS and authenticates by one-time passcode. No password is required at any point.

5.2 Daily check-in

ID	Requirement
FR-6	The patient receives a daily reminder at a configured local time. Where push notification is unavailable or unreliable, SMS is used as fallback.
FR-7	The check-in form is composed server-side from the patient's post-operative day. Fields not active on that day are absent, not disabled.
FR-8	Newly activated questions are visually marked as new on their first appearance.
FR-9	All binary questions share the same polarity: "yes" is always the concerning answer.
FR-10	Binary questions default to unanswered. No default answer may be pre-selected.
FR-11	The patient can capture and upload a wound photograph, retake it before submitting, and see the previous day's photo for comparison.
FR-12	A submission made without connectivity is queued locally and transmitted when connectivity returns, preserving the original capture time.
FR-13	A submission is accepted with missing fields. Missing values are recorded as unknown and never treated as normal.
FR-14	The patient never sees a risk score, tier or alert status. The only risk content shown is emergency guidance under FR-22.

5.3 Data captured

Item	Type	Active	Purpose
Temperature	Numeric °C	Day 0–42	Infection
Pulse	Numeric bpm	Day 0–42	Infection, clot, bleeding
Oxygen saturation	Numeric %	Day 0–14	Clot on the lung
Pain score	Scale 0–10	Day 0–84	Pain control, general
Steps walked	Numeric	Day 0–84	Recovery trajectory
Wound photograph	Image	Day 0–14	Infection
Calf pain or swelling	Binary	Day 0–42	Clot in the leg
Breathlessness or chest pain	Binary	Day 0–42	Clot on the lung
Wound still leaking	Binary	Day 5–28	Deep infection
Redness spreading	Binary	Day 0–28	Wound infection
Shivering or chills	Binary	Day 0–28	Systemic infection
Harder to bear weight	Binary	Day 0–42	Fracture, infection, mechanical
Knee bend proxy	Chair / car / bike	Day 14–84	Range of motion milestone
Can lay leg flat	Binary	Day 14–84	Extension milestone
Walking aid	Walker / cane / none	Day 14–84	Function milestone
Blood thinner taken	Binary	Day 0–35	Adherence, clot prevention

> **ON THE RED-FLAG QUESTIONS**
>
> Questions comparing against something ("redness spreading wider than yesterday", "calf more swollen than your other leg") must carry that comparison in the wording. Without the anchor, nearly every post-operative patient answers yes, because the operated knee is legitimately swollen and the incision is legitimately pink.

5.4 Cadence

Period	Frequency	Change
Day 0 to 14	Daily	Full safety set including wound photograph
Day 15 to 42	Every three days	Oxygen and photograph drop off; progress section activates
Day 43 to 84	Weekly	Milestone tracking predominates
Day 84	Programme ends	Final review, patient marked complete

ID	Requirement
FR-15	A cadence change is announced to the patient on the check-in immediately preceding it.
FR-16	A missed expected check-in raises its own distinct status, visible on the dashboard and separate from any clinical tier.

5.5 Risk evaluation

ID	Requirement
FR-17	Every submission is evaluated immediately on receipt and assigned exactly one tier: normal, review, critical or emergency.
FR-18	Evaluation combines threshold breaches, day-indexed expected ranges, and trend deviation from the patient's own recent history.
FR-19	Any single signal reaching its maximum severity escalates the evaluation regardless of the overall score. A good total may not dilute a critical individual finding.
FR-20	A combination of moderate findings that individually clear their thresholds may still escalate on the weighted total.
FR-21	Every evaluation records the ruleset version that produced it, the contributing signals, and their ranked contribution.
FR-22	A reported breathlessness or chest pain triggers emergency tier, displays emergency guidance to the patient immediately, and notifies the care team in parallel. It does not wait in any queue.

5.6 Alerting and escalation

ID	Requirement
FR-23	Critical and emergency tiers create an alert and notify the on-call nurse by push notification and SMS immediately.
FR-24	Review tier places the patient in the dashboard queue without an interrupting notification.
FR-25	A critical alert unacknowledged within a configurable window, defaulting to fifteen minutes, escalates automatically to a backup nurse or the on-call physician.
FR-26	Repeat alerts arising from the same set of reasons are suppressed for twenty-four hours unless the tier increases.
FR-27	Every alert must reach a recorded outcome. Alerts cannot be dismissed without one.

5.7 Care team workflow

ID	Requirement
FR-28	The dashboard lists patients ordered by tier and then by alert age. The ordering is server-determined; the user makes no triage decision.
FR-29	Each row states the specific reasons in clinical language. The numeric score is never displayed.
FR-30	A patient detail view shows trends for each tracked parameter against the discharge baseline, plus the wound photograph series.
FR-31	Opening an alert presents the question script associated with the rule that fired.
FR-32	Resolution requires selecting one of four outcomes: reassured and continuing, care plan adjusted, same-day review arranged, or directed to emergency care. A free-text note is optional.
FR-33	Selecting the emergency outcome additionally notifies the surgical team.

SECTION 6

Non-functional requirements

ID	Area	Requirement
NFR-1	Evaluation latency	A submission is evaluated and acknowledged to the device within two seconds at the 95th percentile, because emergency guidance depends on the synchronous result.
NFR-2	Notification latency	A critical alert dispatches to the on-call nurse within sixty seconds of submission receipt.
NFR-3	Availability	99.5% monthly for the check-in and alerting path. The dashboard may degrade before ingest does.
NFR-4	Scale assumption	Up to 500 concurrently enrolled patients, one submission each per day. Write volume is negligible by design; no message broker or cache is warranted.
NFR-5	Accessibility	Minimum 16px body text, minimum 44px touch targets, contrast meeting WCAG AA. Usable one-handed while seated.
NFR-6	Localisation	English, Tamil and Hindi at launch, professionally translated rather than machine translated. Anatomical references accompanied by a diagram.
NFR-7	Device support	Android 9 and above, Chrome and Safari current and one prior major version. Functional on 3G.
NFR-8	Offline tolerance	Up to seven days of queued submissions retained locally without loss.
NFR-9	Auditability	Every evaluation, alert, notification and outcome is immutably recorded with actor and timestamp, retained for the statutory period.
NFR-10	Determinism	Re-evaluating a stored reading with the same ruleset version must reproduce the original tier and reasons exactly.

Security and data protection

ID	Requirement
NFR-11	Transport encryption throughout; encryption at rest for the database and photograph storage.
NFR-12	Wound photographs are served only through short-lived signed URLs, never public paths.
NFR-13	Patient authentication by one-time passcode. Clinician authentication with a second factor and role-based access.
NFR-14	Explicit versioned consent captured at enrolment, with purpose limitation recorded and withdrawal supported.
NFR-15	A documented breach process capable of notifying the Data Protection Board and affected patients within seventy-two hours.

REGULATORY TIMING

Health data places the product squarely within the Digital Personal Data Protection regime. The DPDP Rules were notified on 13 November 2025, with the Consent Manager framework effective 13 November 2026 and the principal substantive provisions from 13 May 2027. There is runway, but consent capture, purpose limitation and the breach process must be built into the first version. Retrofitting consent architecture later is materially harder than building it now.

SECTION 7

Success metrics

Metric	Target	Why it matters
Daily check-in completion	≥70% in first two weeks	The system is inert without submissions. This is the single best proxy for whether the form is tolerable.
Critical alert acknowledgement	≥90% within 15 minutes	Measures the human half of the loop. A low figure means the rostering is wrong, not the software.
Alerts per nurse per day	≤5 in steady state	The alert fatigue guardrail. Above this, thresholds are miscalibrated and the queue will be ignored.
Proportion of alerts resolved with an outcome	100%	Unresolved alerts accumulate and destroy trust in the queue.
Escalations to the surgeon	Tracked, not targeted	A rising figure indicates the nurse coverage window is inadequate.
Time from complication onset to clinical contact	Baselined in pilot	The outcome the product exists to improve. Requires chart review to establish.

What is deliberately not a launch metric

Thirty-day readmission rate. It is the outcome that ultimately matters and the number a hospital will eventually buy on, but at pilot scale it will not reach significance and optimising against it early encourages over-alerting.

SECTION 8

Delivery and risks

Engineering estimate

Approximately 68 engineer-days of build, assuming two senior engineers working with an AI coding assistant. Split across two people with imperfect parallelism, that is seven to nine calendar weeks to a feature-complete deployable MVP, with a clickable demo on seeded data available from around week three.

Workstream	Eng-days	Workstream	Eng-days
Schema, migrations, seed data	2	Patient app form and accessibility	6
Patient OTP and clinician auth	6	Photo capture, compression, upload	4
Check-in ingest and validation	3	Offline queue and retry	5
Rules engine and test suite	4	Reminders and SMS fallback	4
Alert service, dedup, escalation	6	Dashboard list, detail, trends	7
SMS integration	2	Acknowledge and outcome flow	3
Dashboard queries and sorting	3	Infra, CI, storage, monitoring	5
		Integration, debugging, polish	**8**

> **THE DATE IS NOT SET BY ENGINEERING**
>
> Roughly four months to first real patient, not eight weeks. The critical path runs through surgeon sign-off on threshold values, rostering a nurse to answer alerts, two rounds of usability testing with patients in the target age band, and consent and breach processes. All four should start on day one, in parallel with the build.

Recommended sequencing

Do not build the patient application first. Build the rules engine and clinic dashboard in weeks one to three, and populate it with data a nurse collects by telephone from real discharged patients. Run that for two weeks. It is inelegant and does not scale, but it establishes whether the risk logic flags the right people and whether the queue is useful to a nurse, which is the assumption the entire product rests on. If the thresholds are wrong, that is discovered for three weeks of effort rather than three months.

Principal risks

Risk	Severity	Mitigation
Alert fatigue from over-triggering	High	Two tiers, day-indexed bands, deduplication, hysteresis, and an explicit alerts-per-nurse-per-day guardrail.
No nurse rostered to answer alerts	High	Named on-call coverage is a contractual deployment prerequisite, not a configuration option.
Poor patient adherence	High	Declining cadence, minimal field count, SMS fallback, family-member operability, usability testing in the target age band.
Thresholds unapproved or contested	High	Configuration-driven rulesets, versioned and stamped on every decision; synthetic fixture patients demonstrating first-escalation day for clinical review.
Missed complication despite monitoring	Severe	Patient-side emergency guidance independent of acknowledgement; escalation timer; explicit product limitations in patient onboarding.
Consent architecture retrofit	Medium	Built into the first version rather than deferred.

Open questions

1. Who holds on-call responsibility outside clinic hours, and what is the committed coverage window?
2. Does the pilot clinic supply vitals devices, or does the patient use what they already own?
3. What is the fallback when a patient has no smartphone but a family member does?
4. Where does the same-day review outcome route in the clinic's existing booking process?
5. Is the product name settled, and has trademark clearance been obtained across classes 9, 42 and 44?